

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

M.Tech. (EC)(C.S.E.) SEM – I Examination, 2009-10

EC - 703 Digital System Design
Date: 26.12.09, Saturday Time: 1.00 p.m to 4.00 p.m Maximum Marks:70

Instructions:

1. Attempt all questions.
2. Answer to the two sections must be written in separate answer books.
3. Figures to the right indicate *full* marks.
4. Make suitable assumptions and draw neat figures wherever required.

SECTION-I

Q-1(a) A clocked mealy sequential network with one input (X) and one output (Z) is to be designed. The output is to be 0, unless the input is 0 following a sequence of exactly two 0 inputs followed by a 1 input. Specify the output sequence for each of the following input sequences: 3

(a) X=....10010 (... means any input sequence not ending in 00)

Z=_____

(b) X=...00010

Z=_____

(c) X=00100100010

Z=_____

(b) Differentiate between inertial delay model and Transport delay model in terms of signal assignments. 3

(c) Draw a block diagram of a 4 bit serial adder with Accumulator. 3

(d) Explain the different components of SM Chart. 2

Q-2 (a) Design a 4X4 array multiplier and explain it in detail. 5

(b) Give a VHDL description of a 4 bit carry ripple adder. 4

(c) Explain the importance of setup and hold time in logic designing. 3

OR

(a) Design a divider along with control graph for signed (2's complement) binary numbers that divides a 32 bit dividend by a 16 bit divisor to give a 16 bit quotient. 7

- (b) (i) Draw a schematic diagram for an LFSR with $n=4$ that generates maximum length sequence. 4
(ii) Add logic so that 0000 is included in the state sequence.
(iii) Determine the actual state sequence.

OR

- (a) Explain a 4-bit BILBO register. How the BILBO registers are used for testing a network with two combinational blocks? 8
(b) Explain self test circuit for RAM using signature register. 4

- Q-6 (a) Draw and explain the SM chart for UART transmitter. 6
(b) The following stated table is implemented using ROM and two D flip-flop.(negative edge triggered). 6

Q_1 Q_2	Q_1^+ Q_2^+		Z	
	$x=0$	$x=1$	$x=0$	$x=1$
00	01	10	0	1
01	10	00	1	1
10	00	01	1	0

- (i) Draw the block diagram.
(ii) Write VHDL code that describes the system.

OR

- Q-6 (a) Compare the different programming technologies used in PLD. 6
(b) Which are the main modules of keypad scanner? Explain the working of keyscan using state graph. 6

(3)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

M.Tech. (EC) (C.S.E.) Sem. - I Examination, 2009-10
EC – 704 Digital Signal Processing & Applications

Date: 28/12/2009, Monday

Time: 1:00 p.m to 4:00 p.m

Maximum Marks:70

Instructions:

1. Attempt all questions.
2. Answer to the two sections must be written in separate answer books.
3. Figures to the right indicate *full* marks.
4. Make suitable assumptions and draw neat figures wherever required.

SECTION -I

Q-1(a) The analog signal given below is sampled by 600 samples per second 3

$x(t) = 2\sin 480\pi t + 3\sin 720\pi t$, Calculate:

- 1) Nyquist sampling rate and maximum frequency
- 2) What are the frequencies in the radians in the resulting discrete signal $x(n)$?

(b) Give Difference equation representation of Moving-Average system. 4
Find the frequency response of the Moving-Average system.

(c) Determine Impulse response for a stable linear time-invariant system 3
for which the input $x[n]$ and output $y[n]$ satisfy the linear constant-coefficient difference equation

$$y[n] - 0.5y[n-1] = x[n] - 0.25x[n-1]$$

(d) Sketch the R.O.C. of delayed unit impulse sequence $\delta[n-7]$. 1

Q-2(a) Consider the three sequences 5

$$v[n] = u[n] - u[n-6],$$

$$w[n] = \delta[n] + 2\delta[n-2] + \delta[n-4],$$

$$q[n] = v[n] * w[n]$$

(a) Find and sketch the sequence $q[n]$.

(b) Find and sketch the sequence $r[n]$ such that $r[n] * v[n] = \sum_{k=-\infty}^{n-1} q[k]$